MODULE 1

Introduction to Hydrogen

Properties · energy carrier · colours · demand

Learning objectives

- State the key physical and combustion properties of hydrogen and what they mean for engineering design.
- Explain why hydrogen is an **energy carrier**, not a primary energy source.
- Compare hydrogen with electricity, natural gas and diesel on energy per kg and per litre.
- Classify hydrogen by production route (colour codes) and by emissions intensity.
- Describe where hydrogen is used today and where new demand is expected.

1.1 Basic concept

Hydrogen (H_2) is the lightest molecule. It does not occur free in useful quantities on Earth, so it must be **produced** from a hydrogen-containing compound – mainly water (H_2O) or hydrocarbons such as methane (CH_4) and coal. Energy is spent to make it, and a part of that energy is recovered when hydrogen is burned or used in a fuel cell. That is why hydrogen is described as an **energy carrier** – like electricity – rather than an energy source.

1.2 Engineering properties you will use

Property	Value	Engineering meaning
Molar mass	2.016 g/mol	Very light; leaks easily through small gaps and permeates some materials
Gas density (0 °C, 1 atm)	0.0899 kg/m³ (≈1/14 of air)	Rises and disperses quickly outdoors; accumulates under roofs indoors
Normal boiling point	–252.9 °C (20.3 K)	Liquid H_2 needs cryogenic, vacuum-insulated equipment
Liquid density	≈71 kg/m³	Even liquid H_2 is ~10× lighter than diesel – volume is the challenge
Lower heating value (LHV)	120 MJ/kg = 33.3 kWh/kg	Basis for fuel-cell / engine efficiency
Higher heating value (HHV)	142 MJ/kg = 39.4 kWh/kg	Basis for electrolyser efficiency (water leaves as liquid)
Flammability range in air	4 – 75 vol %	Very wide range – leaks must be detected and ventilated
Minimum ignition energy	≈0.017 mJ (in air)	Very easy to ignite – control static and ignition sources
Auto-ignition temperature	≈585 °C (858 K)	Gas group IIC, temperature class T1 in hazardous-area design

Values: Stolten (ed.), *Hydrogen and Fuel Cells*, Wiley-VCH, ch. 1 [R6]; LHV/HHV kWh conversion used by IRENA [R1].

1.3 Hydrogen versus other energy carriers

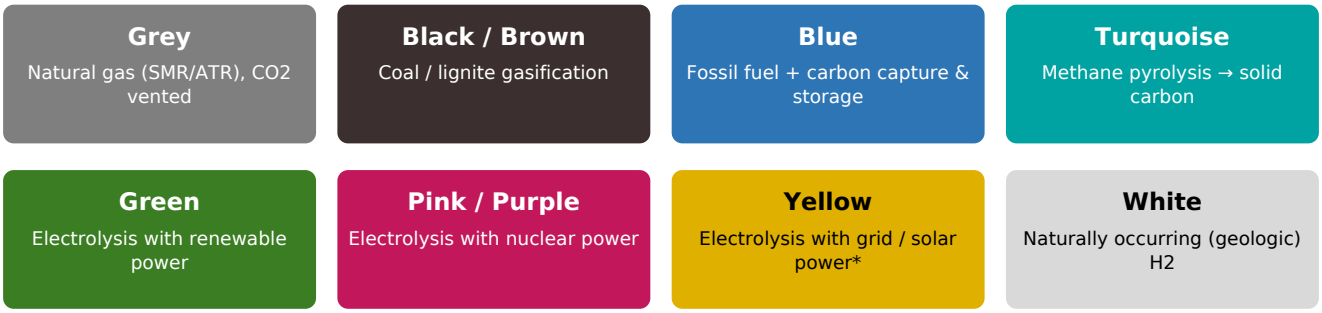
Carrier	Energy per kg	Energy per volume	Key point
Hydrogen	Highest (33.3 kWh/kg LHV)	Lowest – needs compression, liquefaction or a carrier molecule	Excellent per kg, poor per litre
Natural gas (CH_4)	Lower than H_2	≈3× higher than H_2 gas at the same pressure	Existing pipelines and know-how

Carrier	Energy per kg	Energy per volume	Key point
Diesel / petrol	Lower than H_2	Very high – liquid at room temperature	Easy storage; CO_2 on combustion
Electricity (battery)	Low per kg of battery	Medium	Highest round-trip efficiency; no molecule to store

Engineering principle – why conversion losses matter

Every conversion step loses energy: electricity → electrolysis → compression or liquefaction → transport → fuel cell → electricity. Where a direct electric solution exists (e.g. a battery car on short duty), it usually wins on efficiency. Hydrogen is most valuable where you need a **molecule** (ammonia, refining, steel reduction), **long-duration storage**, or **high energy per kg** (heavy, long-haul transport).

1.4 Hydrogen classifications (colour codes)



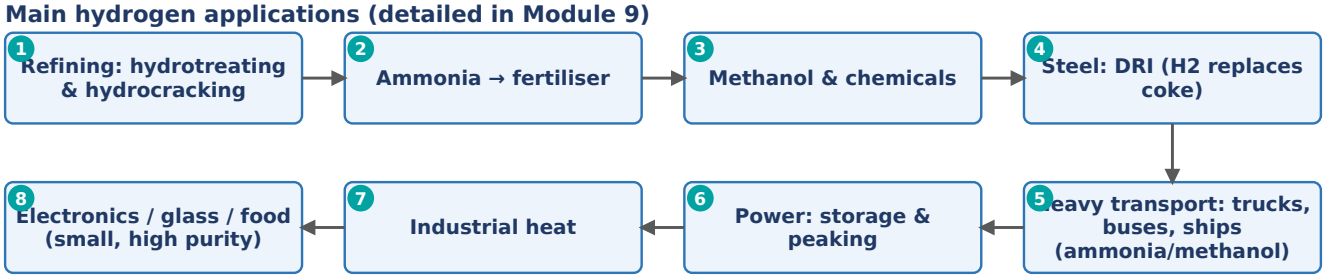
*Colour names are informal and vary by country (*yellow is used for both solar and grid electricity). The IEA recommends classifying hydrogen by **emissions intensity** (kg CO_2-eq per kg H_2) instead [R5].*

Production route	Emissions intensity (kg CO_2-eq / kg H_2)	Source
Unabated natural gas (SMR)	10 – 14 (of which 1 – 5 upstream/midstream)	IEA [R5]
Natural gas + CCS, ~60 % capture	≈5 – 8	IEA [R5]
Natural gas + CCS, >90 % capture	0.8 – 6	IEA [R5]
Unabated fossil (worst cases, incl. coal)	up to ≈27	IEA [R5]
Electrolysis with renewable electricity	0 (IPHE method, production stage)	IEA [R5]
Global average production (2021)	12 – 13	IEA [R5]
India Green Hydrogen Standard threshold	≤ 2 (well-to-gate, 12-month average)	MNRE, Aug 2023 [R10]

1.5 Demand and applications

Global hydrogen demand was **above 97 Mt in 2023** [R3] and **almost 100 Mt in 2024**, up 2 % [R4]. Demand is concentrated in **oil refining** and **industry** (ammonia, methanol, steel via direct reduction). New uses – heavy industry, long-distance transport, energy storage – were still **below 1 %** of demand in 2023, and low-emissions hydrogen production was under 1 Mt [R3]. China is the largest producer of fuel-cell commercial vehicles (≈95 % of world stock) [R4].

Main hydrogen applications (detailed in Module 9)



1.6 Engineering challenges

- **Cost** – low-emissions hydrogen is still more expensive than unabated fossil hydrogen; the gap is expected to narrow [R4].
- **Storage density** – very low energy per litre at ambient conditions.
- **Materials** – embrittlement of high-strength steels; permeation and leakage.
- **Safety** – wide flammability range and very low ignition energy; invisible flame in daylight.
- **Infrastructure** – pipelines, refuelling stations and certification schemes are still being built.

Quiz & revision – Module 1

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Why is hydrogen called an energy carrier?
a) It is found free in nature **b)** It must be produced using another energy source **c)** It is a fossil fuel **d)** It cannot be burned

Q2. Which heating value is used for electrolyser efficiency?
a) LHV 33.3 kWh/kg **b)** HHV 39.4 kWh/kg **c)** Either – they are equal **d)** Neither

Q3. Flammability range of H2 in air:
a) 5–15 % **b)** 1–8 % **c)** 4–75 % **d)** 20–40 %

Q4. IEA recommends classifying hydrogen by:
a) Colour **b)** Price **c)** Emissions intensity **d)** Purity

Q5. Largest current use of hydrogen:
a) Passenger cars **b)** Refining & industry **c)** Aviation **d)** Home heating

Answer key
Q1 – b: it must be made from water or hydrocarbons; **Q2 – b**: water leaves as liquid in the reference reaction; **Q3 – c**: 4–75 vol %; **Q4 – c**: kg CO2-eq per kg H2; **Q5 – b**: refining, ammonia, methanol, steel